



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2028	M.TECH Dual Degree 5Y	IIT Kharagpur	8.65 / 10
2023	MPBSE CLASS XII	Alpha Higher Secondary School	94%
2021	MPBSE CLASS X	Alpha Higher Secondary School	100%

INTERNSHIPS

Physics-Guided Reservoir Inflow Forecasting | Research Intern | Indian Institute of Technology Kharagpur [Jun '25 - Present]
Advisor: Prof. Chandranath Chatterjee, Indian Institute of Technology Kharagpur

- Implemented a **ResNet**-based CNN for ECMWF S2S rainfall downscaling, achieving RMSE **5.60** mm/day and R2 **0.873**
- Evaluated performance across **2,108** test samples (**52,700** predictions), achieving R2 **0.776** on heavy rainfall events
- Developing a **VIC** hydrological and **Transformer**-based **FutureTST** framework for reservoir **7-30-day** inflow forecasting

PROJECTS

Rare Skin Cancer Detection via Generative Augmentation and Explainable AI | Self Project [Nov '25 - Jan '26]

- Constructed a lesion-level stratified split for **ResNet18** on the **10K-image HAM10000** dataset, preventing data leakage
- Developed a **TTUR**-stabilized **DCGAN**, generating **400** diverse synthetic **Dermatofibroma** samples from only **71** images
- Conducted **ablation experiments** on weighted loss, classical and **generative augmentation** under a **70:1** imbalance
- Attained **0.8166** Macro F1, with **Grad-CAM** confirming **lesion-focused** attention while ignoring background artifacts

Discrete Visual Representation Learning | Self Project [May '26 - Jun '26]

- Implemented a **VQ-VAE** with **EMA** quantization and **commitment** loss, learning **512** tokens from **202K** CelebA images
- Compressed **64x64** RGB faces into **88** discrete token grids, achieving **100%** codebook utilization without collapse
- Trained a **6-layer** Mini-GPT Transformer with **causal self-attention** on **202K** image latents and a **513-token** vocabulary
- Reduced **Transformer** cross-entropy to **3.14** through autoregressive visual token modeling and image generation

COMPETITION/CONFERENCE

Agentic Voice Support System with RAG and Sentiment-Aware Escalation | Airia AI Hackathon [Feb '26 - Apr '26]

- Developed a **multilingual voice agent** using FastAPI, WebSockets, and Sarvam STT/TTS for seamless real-time support
- Implemented **OpenAI** Function Calling with **phonetic normalization** for routing across **5** customer support intents
- Built a hybrid **FAISS-RAG** and **SQLite** retrieval layer, combining policy retrieval with transactional support workflows
- Architected a **sentiment-aware escalation** engine using session memory, triggering handoffs beyond **70%** negativity

Long Narrative Consistency Verification System | KDSH | IIT Kharagpur [Dec '25 - Jan '26]

- Architected a deterministic **GraphRAG** pipeline for verifying **100K+** word narratives using **rule-based reasoning** engine
- Designed an **agentic workflow** for claim decomposition, probe-question generation, and evidence-set verification
- Integrated hybrid retrieval by fusing Knowledge Graph, **ChromaDB**, and **BM25** using **Reciprocal Rank Fusion** ranking
- Delivered **71.0%** verification accuracy and **0.60** Macro-F1 with an **Evidence-Set Verifier** for long-context verification

AWARDS AND ACHIEVEMENTS

- Ranked **17th** globally in the **Airia AI Agents Hackathon**, competing against **1,491** international participant teams
- Qualified the Institute **Internal Hackathon** and was nominated by IIT Kharagpur for **Smart India Hackathon 2025**
- Achieved a ranking within the top **10%** among **1,50,000+** qualified candidates in the Joint Entrance Exam 2023

SKILLS AND EXPERTISE

Programming Languages: C++ | Python | C | SQL* (**elementary proficiency*)
Technologies: LangChain | LangGraph | Hugging Face Transformers | FastAPI | Docker | MySQL | Streamlit | GitHub
Libraries: PyTorch | Scikit-learn | Pandas | NumPy | Matplotlib | Seaborn | Keras | spaCy | Pydantic | C++ STL

COURSEWORK INFORMATION

Coursework: Programming and Data Structures | Probability and Statistics | Advanced Calculus | Linear Algebra and Complex Analysis | Theory of Partial Differential Equations | Economics | Basic Electronics
MOOCs: Deep Generative Modelling Stanford Online | Summer Analytics, Consulting & Analytics Club, IIT Guwahati

POSITIONS OF RESPONSIBILITY

Mess Secretary | MMM Hall of Residence, IIT Kharagpur [Jul '24 - Apr '25]

- Managed dining operations and food quality for **2000+** residents through efficient staff and student coordination

EXTRA CURRICULAR ACTIVITIES

- Contributed to the **Illumination** team of MMM Hall, which secured **1st position** among **20+** halls in cultural event
- Served as **wing representative**, coordinating activities and welfare initiatives for a community of **100+** wing members